Two Reports, One Day: Do Investors Neglect Bundled Earnings News?

Draft, October 2026

Abstract

Chinese listed firms must publish their annual report and their first-quarter (Q1) report by 30 April, and the Q1 report may not appear before the annual report. As annual reports drifted toward the deadline, the share of firms that release both reports on the same day rose from 4% to 47% between 2006 and 2020. Such bundles give a clean, within-firm test of limited attention: two earnings signals of the same kind arrive at once. A naive comparison suggests that investors neglect the older signal. The stock-price response to the annual earnings surprise is 52% lower when the Q1 report arrives in the same month. We show that this gap is an artefact of comparing different information sets. Once both reports are public for every firm, the response to the annual surprise is the same. Comparing firms that filed their annual report on the same day, and letting each filing date have its own earnings response, bundling changes neither the price response to either surprise nor the later drift. Trading volume reacts, if anything, more strongly to bundled news. Bundling is mostly a mechanical by-product of late annual reporting, and firms do not use it to bury bad annual news. Muted reactions to bundled disclosures should not be read as inattention without aligning what investors know.

Keywords: limited attention, disclosure bundling, earnings announcements, information overload, China. **JEL:** G14, G41, M41.

1 Introduction

Attention is scarce. Investors cannot read everything at once, and a large literature argues that this limits how quickly prices absorb public information (Kahneman, 1973; Sims, 2003; Hirshleifer and Teoh, 2003; Peng and Xiong, 2006; Blankespoor et al., 2020). If attention is scarce, then *how* a firm packages its news should matter. Two pieces of news released together compete for the same few hours of investor attention; released a week apart, each gets its own. This concern is behind evidence that concurrent releases of earnings and the 10-K receive muted reactions and more drift (Arif et al., 2019), that managers spread disclosures out to avoid overload (Chapman et al., 2019), and that managers bundle or time bad news when attention is low (deHaan et al., 2015; Segal and Segal, 2016; Michaely et al., 2016).

Testing whether packaging *causes* neglect is hard. Firms choose their packages. The pieces of a typical bundle, such as an earnings release with a management forecast, are different kinds of news, which makes it difficult to measure the surprise in each (Rogers and Van Buskirk, 2013). And the reactions of bundled and unbundled news are compared at moments when investors know different things.

We study a setting that removes most of these problems. In China, two rules create bundles. Both the annual report and the Q1 report are due by 30 April, and the Q1 report may not be published before the annual report (CSRC, 2021; Shanghai Stock Exchange, 2023). A firm

that files its annual report on 26 April must therefore publish its Q1 report between 26 and 30 April, often on the same day (Figure 1). As annual reports moved toward the deadline, same-day bundles rose from 4% of firm-years in 2006–2011 to 47% in 2020. Each bundle contains two earnings numbers of the same kind, the fourth-quarter result inside the annual report and the next Q1 result, which we measure with the same expectation model.

A naive look at the data suggests textbook neglect. In the month of the annual report, the stock-price response to the annual earnings surprise is 0.49% per unit of surprise when the Q1 report comes in a later month, but only 0.24% when both reports come out on the same day: a 52% drop ($t = -3.41$). We argue that this comparison is wrong, for two reasons, and design the tests around them.

The first reason is *information sets*. When the Q1 report comes a month later, investors price the annual surprise without knowing Q1. A good annual result predicts a good Q1, so part of the price response to the annual surprise is really a forecast of Q1. When both reports arrive together, that forecast is replaced by the actual Q1 number, and the price response is rightly attributed to the Q1 surprise. An efficient market therefore reacts less to a bundled annual surprise, without any inattention. Once we compare returns over a window in which both reports are public for every firm, the gap disappears: 0.25% versus 0.25% ($t = -0.03$).

The second reason is *selection*. Bundlers are late filers with worse news. Our main design therefore compares a bundling firm only with firms that filed their annual report *on the same day* but published Q1 a few days later in the same month. By the end of that month the two groups have released exactly the same kind of information. We let each filing date have its own response to both surprises, so that any news, crowding or lateness that is common to the date cancels out. In other specifications we let each firm have its own response, comparing the same firm in years it bundles with years it does not. Across these designs bundling has no effect on the price response to either surprise, and no effect on the drift over the next three months. For the Q1 surprise we can rule out that bundling lowers the response by more than 21% of its normal size. Trading volume, an independent measure of attention, does not respond less to bundled news; it responds more to the size of a bundled Q1 surprise ($t = 2.12$).

Who bundles, and why? The lateness of the annual report explains most of it: each decile of lateness raises the probability of a same-day release by 7.3 percentage points. Holding the filing date fixed, firms with bad annual news are not more likely to bundle, so bundling is not used to bury bad annual news, and bundled bad news is not priced less. Firms with bad Q1 news are *less* likely to bundle, consistent with the older finding that bad news is delayed (Givoly and Palmon, 1982; Haw et al., 2000).

The paper makes three contributions. First, it provides clean evidence that releasing two earnings signals on the same day does not cause investors to neglect either of them, at least at a monthly horizon. This complements the evidence that macro news makes investors process firm news *better* (Hirshleifer and Sheng, 2022), and the finding of Segal and Segal (2016) that strategically bundled news shows no sign of investor inattention. It qualifies the interpretation of muted reactions to concurrent releases (Arif et al., 2019). Second, it shows a simple bias in such comparisons and a simple fix: whenever one signal in a package predicts the other, compare responses over a window in which both are public. Third, it documents China's April reporting crunch and shows how a sequencing rule turns late reporting into bundling, which adds to work on the timing of Chinese annual reports (Haw et al., 2000; You and Zhang, 2011; Li et al., 2020).

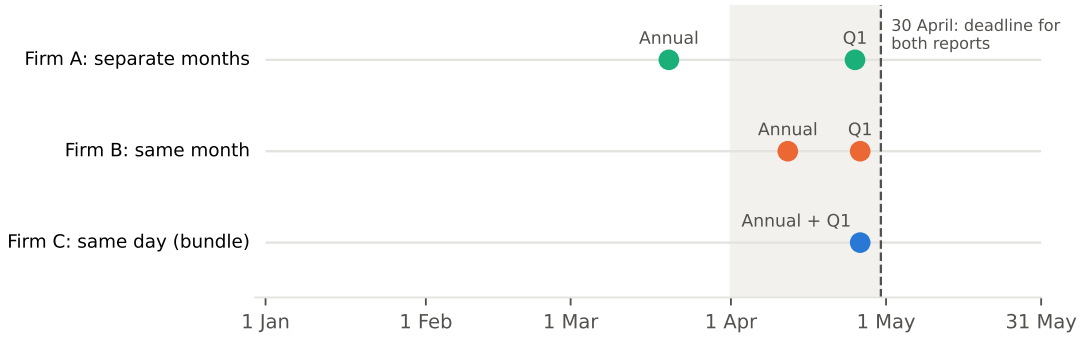


Figure 1: Three ways to publish the annual and the Q1 report. Both reports are due by 30 April, and the Q1 report may not come first. Firm A files its annual report in March and its Q1 report in late April (separate months). Firm B files both in April, a few days apart (same month). Firm C files both on the same day (a bundle). By the end of April all three firms have published the same two reports.

2 Institutional setting and hypotheses

Reporting rules. Chinese A-share firms must publish the annual report within four months of the fiscal year end (31 December), that is, by 30 April, and the Q1 report within one month of the quarter end, also by 30 April. The Q1 report may not be disclosed before the previous year's annual report (CSRC, 2021; Shanghai Stock Exchange, 2023). The exchanges schedule disclosure dates in advance, and many firms crowd into the last week of April. Annual reports give the full-year figures, so the new earnings information they contain beyond the three earlier quarterly reports is the fourth-quarter result. We call the surprise in it the *annual surprise*, SUE^A. The Q1 report gives the first quarter of the new year; we call its surprise the *Q1 surprise*, SUE^Q.

Hypotheses. If attention is limited, a bundle should get less of it per signal (Hirshleifer and Teoh, 2003; Hirshleifer et al., 2009):

- **H1 (neglect).** When the two reports are released together, the price reacts less to at least one of the surprises in the release month, more of the reaction comes later (drift), and trading volume responds less.
- **H0 (full processing).** Given the same information and timing, the package does not matter.
- **H2 (burying).** Managers bundle bad annual news to soften its price impact (deHaan et al., 2015; Segal and Segal, 2016; Li et al., 2020).

H1 does not say *which* signal is neglected. The annual report is the more salient document. It is long, audited and comes with dividends and a full-year headline (Huang et al., 2018). The Q1 number is the more recent and, for forecasting, more relevant one; people tend to weight the most recent evidence more (Hogarth and Einhorn, 1992), and older information moves prices less (Tetlock, 2011). Our tests are symmetric and allow either signal to be neglected.

3 Data

Sample. We use monthly data on all stocks traded on the Shanghai and Shenzhen exchanges from Thomson Reuters Datastream and Worldscope, including inactive stocks to avoid survival bias (Shumway, 1997). The panel gives prices, total return indices, trading volume, shares outstanding, book values and standalone quarterly earnings per share (EPS) together with each report's publication date. We pair every annual report for fiscal year Y with the Q1 report of $Y + 1$, for $Y = 2005$ to 2019 (reports published in 2006–2020). Table 1 shows how the sample of

24,322 firm-years from 2,812 firms is built. We classify each firm-year by the timing of the two reports: *same day* (4,977), *same month* (6,621; median gap 8 days) and *separate months* (12,724). Publication dates in Datastream can differ from the official filing by a day, because many reports are filed in the evening or at weekends. Spot checks against the filings confirm the classification. Hikvision published its 2017 annual report and 2018 Q1 report together on 21 April 2018, and Industrial Bank presented its 2018 annual and 2019 Q1 results together on 29 April 2019. A one-day error does not affect our monthly tests unless it moves a report across a month boundary, a case our robustness tests cover.

Measuring news. We measure each surprise against a seasonal random walk (SRW): expected EPS equals EPS of the same quarter a year earlier (Foster, 1977; Bernard and Thomas, 1990). The surprise is scaled by the standard deviation of the firm's last eight SRW errors. For example, if Q1 EPS is 0.30 against 0.25 a year earlier and past errors have a standard deviation of 0.05, then $SUE^Q = (0.30 - 0.25)/0.05 = 1$. The SRW has a key advantage here: its expectation does not depend on the order of the releases. A statistical forecast made just before a bundled release cannot use the fourth-quarter number, but one made before a separate Q1 release can. That would make Q1 surprises incomparable across groups. We use an out-of-sample XGBoost forecast (Chen and Guestrin, 2016) as a robustness check.

Returns and volume. Returns are monthly and size-adjusted: each stock's return minus the average return of its size decile in that month. A report published on 26 April therefore shows up in the April return. This sounds crude, but it matters less than it seems: by the end of April, firms that bundled and firms that published the two reports a few days apart have revealed exactly the same information, so their April returns can be compared directly. Abnormal turnover is the log of monthly turnover minus the firm's average over the preceding year.

Descriptive statistics. Figure 2 shows the rise of bundling. Until 2012 fewer than 6% of firm-years were same-day bundles; by 2020 nearly half were (panel a). The rise mirrors a shift of annual reports toward the deadline: the median report came 93 days after the year end until 2015 and 114 days in 2020 (panel b). Bundling is almost entirely a late-filer phenomenon: 2% of firms in the earlier half of each year's filing order bundle, against 59% of the latest fifth (panel c). Table 2 shows that bundlers file their annual report 116 days after the year end, against 83 days for firms that publish in separate months. They also bring worse annual news (mean SUE^A of -0.76 against 0.28) and more losses (18% against 8%). They are similar in size and analyst coverage. Selection on news is therefore a first-order concern.

4 Methodology

The naive test. The obvious test regresses the return in the month of the annual report on the annual surprise and lets the slope differ by release pattern:

$$\begin{aligned} R_{i,m} = \alpha_m + \beta\, SUE_i^A + \gamma_{SM}\, SUE_i^A \times SM_i + \gamma_{SD}\, SUE_i^A \times SD_i \\ + \theta\, SUE_i^Q \times (SM_i + SD_i) + \text{dummies} + \varepsilon_i, \end{aligned} \tag{1}$$

where SM and SD mark same-month and same-day releases, α_m are month fixed effects, and the SUE^Q term captures the Q1 news in months that contain both reports. H1 predicts $\gamma_{SD} < 0$.

Table 1: Sample construction

Step	Firm-years
Annual–Q1 report pairs	32,498
Fiscal years 2005–2019	28,769
Q1 not before annual report	28,474
Both surprises available	24,322
Same day	4,977
Same month	6,621
Separate months	12,724

Firm-years pair the annual report for fiscal year Y with the Q1 report of $Y+1$.

Table 2: Descriptive statistics by release pattern

	Separate months	Same month	Same day
Firm-years	12,724	6,621	4,977
Annual report: days after fiscal year end	83	107	116
Days between annual and Q1 report	36.8	9.7	0.0
Annual report released in April	0.21	0.95	1.00
Annual surprise SUE^A	0.28	−0.10	−0.76
Q1 surprise SUE^Q	0.20	0.02	−0.09
Bad annual news ($SUE^A < 0$)	0.42	0.47	0.53
Bad Q1 news ($SUE^Q < 0$)	0.39	0.45	0.48
Annual loss	0.08	0.12	0.18
Size (log market value)	15.92	15.89	15.94
Analyst coverage	0.51	0.48	0.46
Return, annual-report month (%)	0.55	0.19	−0.20
Abnormal turnover, annual-report month	0.23	0.16	0.14

Means. Surprises are SRW-based and scaled by the SD of the firm's past errors. Abnormal turnover is log turnover minus the firm's average over months −12 to −2.

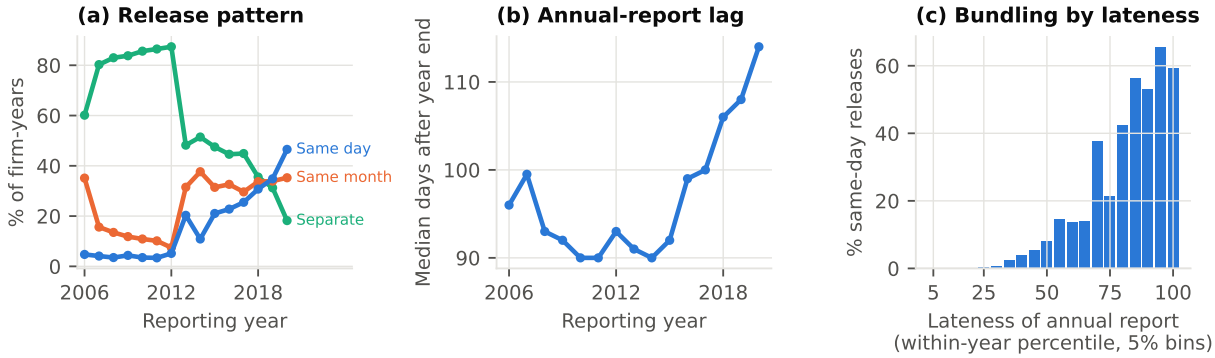


Figure 2: The rise of bundling. (a) Share of firm-years by release pattern. (b) Median number of days between the fiscal year end and the annual report. (c) Share of same-day releases by the lateness of the annual report within its year (percentile of the filing order).

Why the naive test is biased: information sets. Suppose the market is efficient and a one-unit annual surprise predicts a Q1 surprise of ρ units. A firm that reports in separate months first earns $\beta\, SUE^A$ in the annual-report month, where β includes the expected Q1 news. When Q1 arrives, the firm earns $\beta_Q(SUE^Q - \rho\, SUE^A)$, the part of Q1 that was not expected. Adding the two months, the total response to the annual surprise is $\beta - \beta_Q\rho$. A bundling firm reveals both numbers at once, so its single month carries exactly this total: $(\beta - \beta_Q\rho)SUE^A + \beta_Q SUE^Q$. Even an efficient market thus reacts *less* to a bundled annual surprise, by $\beta_Q\rho$. In words, when a strong annual result comes alone, investors also price the strong Q1 they now expect. When the actual Q1 number arrives at the same time, that part of the price move is credited to Q1. The fix is to compare returns over a window in which both reports are public for every firm: from the annual-report month to the Q1-report month.

Main design: same filing date. To deal with selection we compare each bundling firm only with firms that published their annual report on the same day but their Q1 report a few days later in the same month. Within the firm-years whose two reports fall in the same month, we estimate

$$R_{i,m} = \alpha_d + \beta_d\, SUE_i^A + \kappa_d\, SUE_i^Q + \lambda\, B_i + \delta_A\, B_i \times SUE_i^A + \delta_Q\, B_i \times SUE_i^Q + \varepsilon_i, \qquad (2)$$

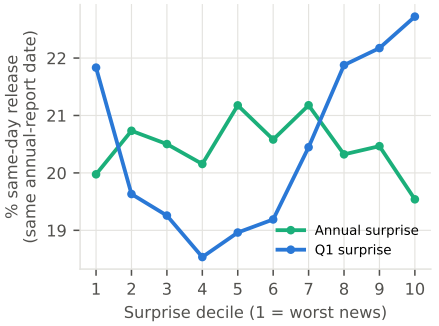
where B_i marks a bundle and d is the annual-report date. Every date has its own intercept *and* its own response to both surprises (β_d, κ_d). For example, on a given day in late April a few dozen firms publish annual reports; some publish Q1 the same day and the rest a few days later.

Table 3: Who bundles?

	(1)	(2)	(3)	(4)
Lateness of annual report (per decile)	7.33***	6.87***		
	(68.63)	(62.83)		
Bad annual news ($SUE^A < 0$)	1.08***	0.69*	0.48	0.36
	(2.61)	(1.73)	(1.39)	(1.08)
Bad Q1 news ($SUE^Q < 0$)	−0.83**	−0.62	−1.36***	−0.99***
	(−1.98)	(−1.53)	(−3.65)	(−2.85)
Annual loss	1.82**	2.30***	−0.26	0.48
	(2.23)	(2.87)	(−0.37)	(0.68)
Size (standardised)	1.38***	2.09***	0.51	0.53
	(4.01)	(4.14)	(1.64)	(1.32)
Analyst coverage	0.43	−0.49	1.21**	0.06
	(0.68)	(−0.79)	(2.10)	(0.11)
Year fixed effects	Yes	Yes	–	–
Annual-report-date fixed effects	–	–	Yes	Yes
Firm fixed effects	–	Yes	–	Yes
Observations	24,322	23,964	24,157	23,800

Linear probability models of a same-day release, in percentage points. t-statistics in parentheses, clustered by firm (columns 1–2) and by firm and annual-report date (3–4). *, **, ***: 10%, 5%, 1%.

Figure 3: Bundling and news, same filing date



Share of same-day releases by surprise decile, after removing annual-report-date effects.

Equation (2) compares only these firms with each other. Anything that is common to the date drops out: market-wide news, the number of other firms reporting that day (Hirshleifer et al., 2009; You and Zhang, 2011), and how late the date is. H1 predicts $\delta_A < 0$ or $\delta_Q < 0$ in the release month, and the opposite sign in later months. In further specifications each firm has its own intercept and slopes, so the comparison is between years in which the *same* firm bundles and years in which it does not (Cunningham, 2021, ch. 8). We estimate the models by partialling out the group-specific intercepts and slopes (the Frisch–Waugh–Lovell theorem, Lovell, 1963). Standard errors are clustered by filing date and by firm (Petersen, 2009; Cameron et al., 2011).

Volume. Prices can be slow even when investors pay attention, and attention can rise without moving prices. We therefore also replace the return with abnormal turnover and the surprises with their absolute values. H1 predicts that turnover responds less to the size of bundled news.

5 Results

5.1 Who bundles?

Table 3 estimates linear probability models of a same-day release. Lateness dominates. Each decile of the filing order raises the probability of a bundle by 7.3 percentage points (column 1), against a base rate of 20%. Without controlling for the filing date, bad annual news predicts bundling (1.1 points, $t = 2.6$). Once firms are compared with others that filed on the same day (column 3), bad annual news no longer matters (0.5 points, $t = 1.4$). Bad annual news predicts bundling only because it predicts late filing (Givoly and Palmon, 1982; Haw et al., 2000). Bad Q1 news, in contrast, makes a bundle *less* likely: it changes the probability by −1.4 points ($t = -3.6$), also within firm (column 4). Figure 3 shows the same pattern without regression assumptions: given the filing date, the probability of a bundle is flat in the annual surprise but rises with the Q1 surprise from the fourth decile upward (only the very worst decile breaks the pattern). Firms that file the annual report late and have good Q1 news publish both together; those with bad Q1 news take the remaining days. This fits delaying bad news, not burying bad annual news.

Table 4: The naive comparison and the information-set benchmark

	(1)	(2)	(3)	(4)	(5)
SUE^A (annual surprise)	0.49***	0.50***	−0.13	0.22***	0.25**
	(7.32)	(7.40)	(−1.58)	(3.94)	(2.49)
SUE^A× Same month	−0.28**				−0.02
	(−2.45)				(−0.23)
SUE^A× Same day	−0.25***				−0.00
	(−3.41)				(−0.03)
SUE^Q (Q1 surprise)			1.12***	1.05***	2.63***
			(5.65)	(9.49)	(9.21)
SUE^Q× Same month or day	1.06***				
	(9.75)				
SUE^Q× Same month					−1.47***
					(−5.62)
SUE^Q× Same day					−1.59***
					(−5.20)
Sample	All	Separate	Separate	Same month/day	All
Return window	Annual month	Annual month	Q1 month	Joint month	Both public
Observations	23,820	12,674	12,675	11,142	23,820

Size-adjusted returns in % per unit of surprise. Column 1 uses all firm-years and the annual-report month. Columns 2–3 use separate-month firms and the annual-report and Q1-report months. Column 4 uses firms whose two reports fall in the same month. Column 5 uses the window from the annual-report month to the Q1-report month. Fixed effects for the calendar month in which the window starts (1–2, 4) or ends (3, 5). t-statistics clustered by month and firm.

5.2 The naive gap and the information-set benchmark

Table 4 column 1 reproduces the naive result. The response to the annual surprise in its release month falls from 0.49% for firms that publish in separate months to 0.21% (same month) and 0.24% (same day). Columns 2–4 trace the information-set argument. For separate-month firms the annual surprise earns 0.49% in its own month (column 2). When Q1 arrives, part of that response is reversed. The annual surprise loads at −0.13% on the Q1-month return, given the Q1 surprise ($t = -1.58$; column 3). The net response once both reports are public is about 0.37%, close to what the simple rule predicts with $\rho = 0.12$ and $\beta_Q = 1.12$ (0.36%). This is already much closer to the bundled firms' 0.22% (column 4). The rest of the difference comes from measuring the two groups over different windows. Column 5 makes the comparison directly, over the window from the annual-report month to the Q1-report month. The annual-surprise slopes are now the same: the bundle interaction is zero ($t = -0.03$). Figure 4 shows the two comparisons side by side.

Column 5 also shows a lower Q1-surprise slope for bundled and same-month firms ($t = -5.20$). This is not evidence of neglect. Separate-month firms have longer windows that start earlier and also absorb information about Q1 before the Q1 report. They are also early filers, a different kind of firm. The same-date design below removes both differences.

5.3 Main result: bundles versus firms filing on the same date

Table 5 reports Equation (2) for the 11,098 firm-years whose two reports fall in the same month, filed on 411 annual-report dates. With date effects only (column 1), the normal release-month responses are 0.19% for the annual and 1.11% for the Q1 surprise. Bundling changes neither: the interactions are 0.04 ($t = 0.58$) and −0.15 ($t = -0.88$). Letting every date have its own response (column 2), every firm have its own response (column 3) or both (column 4) leaves the interactions small and insignificant. Extending the window by one month (column 5), which captures reports

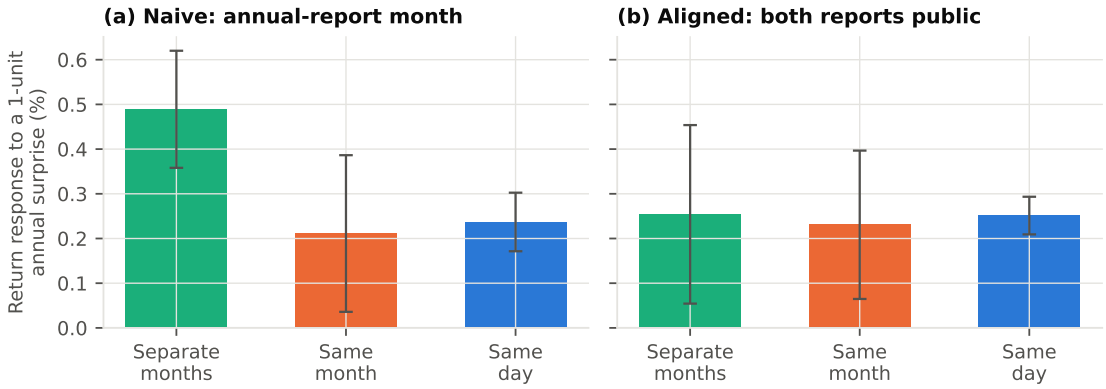


Figure 4: Response to the annual surprise by release pattern. (a) Return in the annual-report month only. (b) Return over the window in which both reports are public. Bars show the slope on SUE^A (% per unit); lines show 95% confidence intervals.

filed in the evening of the last days of April, gives two estimates of opposite sign near the 5% level, a pattern we discuss below. The drift over the next three months (column 6) shows no delayed reaction to either bundled surprise. Bundled firms do earn 1.12% more over these three months whatever their news ($t = 2.12$). This is a level difference between kinds of late filers, not a delayed response to the surprises, and we leave it for future work.

How large an effect can we exclude? In column 2 the 95% confidence interval for the bundle effect on the Q1 response is $[-0.23, 0.60]$, so a reduction of more than 21% of the normal Q1 response is ruled out. The annual surprise moves prices little in its release month to begin with (0.19%), so its interval $[-0.27, 0.09]$ is less informative on its own.

Figure 5 adds the time dimension. It plots the cumulative response to each surprise from January, for bundled and same-month firms filing on the same date. The Q1 paths lie on top of each other before, during and after the release month. The annual-surprise paths differ slightly, but the gap opens in *March*, before either report is out, and stays constant afterwards. That is a difference between the kinds of firms, not an effect of the bundle. The column 5 estimates are not robust. They appear only with date-specific slopes, and in Figure 5 the April–May increments of the two groups are almost identical. Neither the release month nor the next three months shows them. With twelve bundle coefficients in Table 5, one estimate at the 5% level is about what chance alone would produce.

5.4 Trading volume

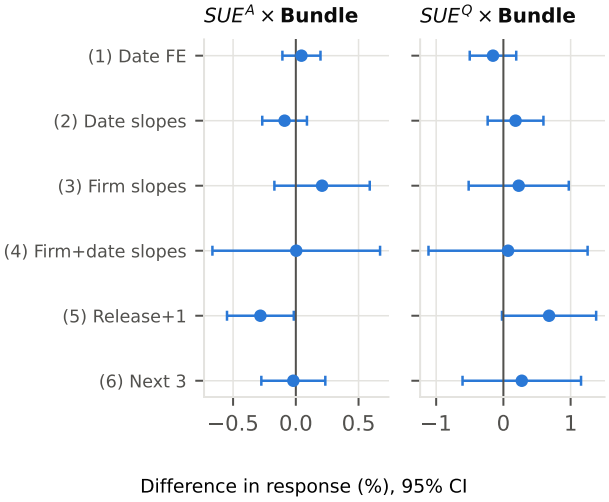
If a bundle diverted attention, fewer investors would trade on its content. Table 6 shows the opposite. Within a filing date, turnover in bundled firms responds *more* to the size of the Q1 surprise (3.21 percentage points per unit of $|SUE^Q|$, $t = 2.12$), and no differently to the size of the annual surprise. The result weakens when each firm gets its own slope (column 4). Figure 6 shows the raw pattern. Bundles look like days of concentrated rather than divided attention, in line with the complementarity of news found by Hirshleifer and Sheng (2022).

5.5 Heterogeneity, robustness, burying and a placebo

Table 7 repeats the main specification (column 2 of Table 5) in subsamples and variations. If attention constraints mattered anywhere, they should matter for small firms and firms without analyst coverage, where fewer investors watch. Neither shows a significant Q1 effect. Small firms

Table 5: Main result: bundled versus same-month firms with the same annual-report date

	(1)	(2)	(3)	(4)	(5)	(6)
SUE^A	0.19***					
	(3.08)					
SUE^Q	1.11***					
	(9.13)					
$SUE^A \times$ Bundle	0.04	−0.09	0.21	0.00	−0.28**	−0.02
	(0.58)	(−0.98)	(1.08)	(0.01)	(−2.08)	(−0.15)
$SUE^Q \times$ Bundle	−0.15	0.18	0.23	0.07	0.68*	0.27
	(−0.88)	(0.85)	(0.60)	(0.11)	(1.90)	(0.61)
Bundle	0.01	−0.09	−0.12	−0.39	0.26	1.12**
	(0.05)	(−0.27)	(−0.23)	(−0.71)	(0.53)	(2.12)
Return window	Release	Release	Release	Release	Release+1	Next 3
Date-specific intercepts	Yes	Yes	Yes	Yes	Yes	Yes
Date-specific slopes	–	Yes	–	Yes	Yes	Yes
Firm-specific slopes	–	–	Yes	Yes	–	–
Observations	11,098	9,212	11,094	10,946	9,211	9,208



Sample: firm-years whose annual and Q1 reports fall in the same calendar month. Bundle = both on the same day. Dependent variable: size-adjusted return (%) in the release month (1–4), the release and the next month (5), or the three following months (6). Date-specific slopes let each annual-report date have its own response to SUE^A and SUE^Q; firm-specific slopes do the same for each firm. t-statistics two-way clustered by date and firm. The plot shows the bundle interactions with 95% confidence intervals.

show a borderline annual effect (-0.27, $t = -1.98$), but no Q1 effect, and the annual effect does not reappear in any other subsample. The results hold before and after 2012, without the 2020 reporting season, when COVID-19 delayed many reports, without loss firms, with XGBoost annual surprises, and over a two-month window. Across 12 rows and two coefficients each, 2 rows contain an estimate at the 5% level, close to the 1.2 that chance alone would produce.

We test burying directly by letting the bundle effect differ for bad annual news. A bundle does not soften the response to bad annual news in the release month (0.06, $t = 0.18$) or over January–July (-1.35, $t = -1.50$). Finally, a placebo assigns "treatment" to same-month firms that will bundle *next* year. If our estimates captured persistent firm types rather than bundling, the placebo would show the same pattern. It does not (-0.29, $t = -0.83$).

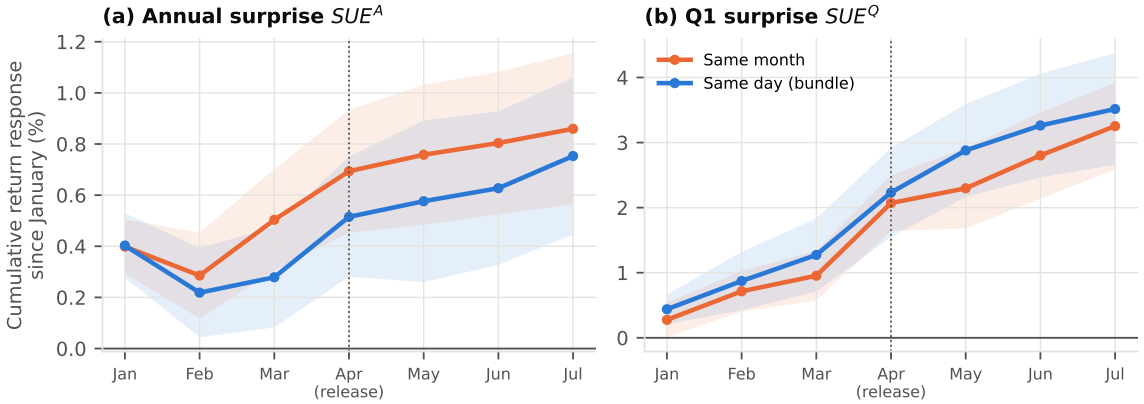


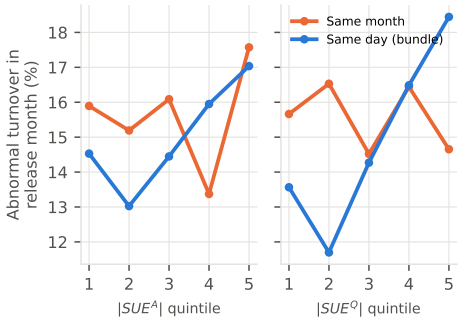
Figure 5: Cumulative response to the two surprises, bundled versus same-month firms. Each point is the slope of the cumulative size-adjusted return from January to the given month on the surprise, from a regression with annual-report date effects. Bands are 95% confidence intervals. Month 0 is the release month (April for 97% of the sample).

Table 6: Trading volume

	(1)	(2)	(3)	(4)
$\lvert SUE^A \rvert$	0.17		0.20	
	(0.44)		(0.45)	
$\lvert SUE^Q \rvert$	−0.03		0.06	
	(−0.04)		(0.06)	
$\lvert SUE^A \rvert \times$ Bundle	0.35	0.12	0.35	−0.32
	(0.71)	(0.17)	(0.56)	(−0.12)
$\lvert SUE^Q \rvert \times$ Bundle	2.27**	3.21**	3.69***	1.90
	(2.09)	(2.12)	(2.84)	(0.34)
Bundle	−3.81*	−4.56*	−4.01	−3.62
	(−1.84)	(−1.94)	(−1.34)	(−0.61)
Date-specific intercepts	Yes	Yes	Yes	Yes
Date-specific slopes	–	Yes	–	Yes
Firm fixed effects / slopes	–	–	FE	Slopes
Observations	10,515	8,775	9,994	10,359

Dependent variable: abnormal turnover in the release month (%). Same sample as Table 5. t-statistics two-way clustered by date and firm.

Figure 6: Turnover and surprise size



Abnormal turnover by quintile of the absolute surprise, after removing date effects.

6 Discussion

Why do bundles not cause neglect? Three features of the setting help. First, a bundle concentrates attention as well as news: the firm's reporting day attracts media coverage, analyst notes and trading, so the second document gets read along with the first. Second, the Q1 report is short and its key number is easy to find, so the processing cost of the second signal is low (Blankespoor et al., 2020). Third, the information that matters for prices is set by the marginal investor, who need not be the distracted one.

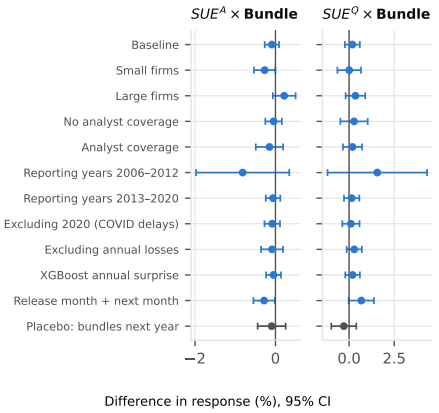
Our evidence has limits. Monthly returns cannot detect delays of a few days. Fedyk (2024) finds that the response to less prominently placed news converges within about two days, which is well inside a month. What we can say is that bundling does not leave news unpriced at the end of the release month or later. Publication dates come from Datastream and can be off by a day. We do not observe Q1 preannouncements, which may tell investors part of the Q1 news before the report; such leaks would affect bundled and same-month firms filing on the same date alike. Finally, the same-date design compares firms that filed the annual report on the same day, and how firms choose their filing date may still be correlated with news. The placebo and the

Table 7: Heterogeneity, robustness, burying and placebo

	$SUE^A \times$ Bundle	$SUE^Q \times$ Bundle	N
Baseline	−0.09 (−0.98)	0.18 (0.85)	9,212
Small firms	−0.27** (−1.98)	0.01 (0.02)	4,220
Large firms	0.22 (1.50)	0.35 (1.26)	4,167
No analyst coverage	−0.05 (−0.44)	0.27 (0.71)	4,299
Analyst coverage	−0.15 (−0.86)	0.19 (0.70)	4,151
Reporting years 2006–2012	−0.82 (−1.38)	1.56 (1.11)	696
Reporting years 2013–2020	−0.06 (−0.66)	0.14 (0.66)	8,516
Excluding 2020 (COVID delays)	−0.08 (−0.82)	0.10 (0.42)	7,757
Excluding annual losses	−0.09 (−0.61)	0.29 (1.33)	7,706
XGBoost annual surprise	−0.05 (−0.51)	0.19 (0.94)	9,212
Release month + next month	−0.28** (−2.08)	0.68* (1.90)	9,211
Placebo: bundles next year	−0.10 (−0.54)	−0.29 (−0.83)	4,880
Burying bad annual news? Extra response to bad annual news in a bundle:			
Release month	0.06 (0.18)		9,126
January–July	−1.35 (−1.50)		9,120

Each row re-estimates column 2 of Table 5. Coefficients in % (t-statistics). Small and large firms split at the yearly median of size. The placebo uses same-month firms only and marks those that bundle in the following year. The burying rows add $SUE^A \times$Bundle$\times$(bad annual news) with date-specific slopes.

Figure 7: Bundle effects across tests



Point estimates and 95% confidence intervals from Table 7; the placebo is grey.

firm-specific slopes address persistent differences, but not time-varying ones.

7 Conclusion

China's reporting rules turned late annual reporting into a natural laboratory for bundled disclosure. A naive comparison suggests that investors largely ignore the annual earnings surprise when the Q1 report arrives with it. The gap disappears once investors' information sets are aligned. In comparisons among firms filing on the same day, bundling changes neither the price response nor the later drift for either signal, and volume reacts more, not less. Firms bundle because they file late, not to bury bad annual news. The broader lesson is methodological: when one signal in a package predicts another, a muted reaction to that signal is what an efficient market should deliver, and only a comparison over an aligned information window can tell neglect from rational updating.

References

Arif, S., Marshall, N. T., Schroeder, J. H., Yohn, T. L., 2019. A growing disparity in earnings disclosure mechanisms: The rise of concurrently released earnings announcements and 10-Ks. *Journal of Accounting and Economics* 68(1), 101221.

Bernard, V. L., Thomas, J. K., 1990. Evidence that stock prices do not fully reflect the implications of current earnings for future earnings. *Journal of Accounting and Economics* 13(4), 305–340.

Blankespoor, E., deHaan, E., Marinovic, I., 2020. Disclosure processing costs, investors' information choice, and equity market outcomes: A review. *Journal of Accounting and Economics* 70(2–3), 101344.

Cameron, A. C., Gelbach, J. B., Miller, D. L., 2011. Robust inference with multiway clustering. *Journal of Business & Economic Statistics* 29(2), 238–249.

Chapman, K., Reiter, N., White, H. D., Williams, C. D., 2019. Information overload and disclosure smoothing. *Review of Accounting Studies* 24(4), 1486–1522.

Chen, T., Guestrin, C., 2016. XGBoost: A scalable tree boosting system. *Proceedings of the 22nd ACM SIGKDD Conference*, 785–794.

China Securities Regulatory Commission, 2021. Measures for the Administration of Information Disclosure by Listed Companies. Beijing.

Cunningham, S., 2021. *Causal Inference: The Mixtape*. Yale University Press, New Haven.

deHaan, E., Shevlin, T., Thornock, J., 2015. Market (in)attention and the strategic scheduling and timing of earnings announcements. *Journal of Accounting and Economics* 60(1), 36–55.

Fedyk, A., 2024. Front-page news: The effect of news positioning on financial markets. *Journal of Finance* 79(1), 5–33.

Foster, G., 1977. Quarterly accounting data: Time-series properties and predictive-ability results. *The Accounting Review* 52(1), 1–21.

Givoly, D., Palmon, D., 1982. Timeliness of annual earnings announcements: Some empirical evidence. *The Accounting Review* 57(3), 486–508.

Haw, I.-M., Qi, D., Wu, W., 2000. Timeliness of annual report releases and market reaction to earnings announcements in an emerging capital market: The case of China. *Journal of International Financial Management & Accounting* 11(2), 108–131.

Hirshleifer, D., Lim, S. S., Teoh, S. H., 2009. Driven to distraction: Extraneous events and underreaction to earnings news. *Journal of Finance* 64(5), 2289–2325.

Hirshleifer, D., Sheng, J., 2022. Macro news and micro news: Complements or substitutes? *Journal of Financial Economics* 145(3), 1006–1024.

Hirshleifer, D., Teoh, S. H., 2003. Limited attention, information disclosure, and financial reporting. *Journal of Accounting and Economics* 36(1–3), 337–386.

Hogarth, R. M., Einhorn, H. J., 1992. Order effects in belief updating: The belief-adjustment model. *Cognitive Psychology* 24(1), 1–55.

Huang, X., Nekrasov, A., Teoh, S. H., 2018. Headline salience, managerial opportunism, and over- and underreactions to earnings. *The Accounting Review* 93(6), 231–255.

Kahneman, D., 1973. *Attention and Effort*. Prentice-Hall, Englewood Cliffs.

Li, T., Xiang, C., Liu, Z., Cai, W., 2020. Annual report disclosure timing and stock price crash risk. *Pacific-Basin Finance Journal* 62, 101392.

Lovell, M. C., 1963. Seasonal adjustment of economic time series and multiple regression analysis. *Journal of the American Statistical Association* 58(304), 993–1010.

Michaely, R., Rubin, A., Vedrashko, A., 2016. Further evidence on the strategic timing of earnings news: Joint analysis of weekdays and times of day. *Journal of Accounting and Economics* 62(1), 24–45.

Peng, L., Xiong, W., 2006. Investor attention, overconfidence and category learning. *Journal of Financial Economics* 80(3), 563–602.

Petersen, M. A., 2009. Estimating standard errors in finance panel data sets: Comparing approaches. *Review of Financial Studies* 22(1), 435–480.

Rogers, J. L., Van Buskirk, A., 2013. Bundled forecasts in empirical accounting research. *Journal of Accounting and Economics* 55(1), 43–65.

Segal, B., Segal, D., 2016. Are managers strategic in reporting non-earnings news? Evidence on timing and news bundling. *Review of Accounting Studies* 21(4), 1203–1244.

Shanghai Stock Exchange, 2023. Rules Governing the Listing of Stocks on the Shanghai Stock Exchange. Shanghai.

Shumway, T., 1997. The delisting bias in CRSP data. *Journal of Finance* 52(1), 327–340.

Sims, C. A., 2003. Implications of rational inattention. *Journal of Monetary Economics* 50(3), 665–690.

Tetlock, P. C., 2011. All the news that's fit to reprint: Do investors react to stale information? *Review of Financial Studies* 24(5), 1481–1512.

You, H., Zhang, X.-J., 2011. Limited attention and stock price drift following earnings announcements and 10-K filings. *China Finance Review International* 1(4), 358–387.

A Variable definitions and replication

Variable	Definition
SUE^A (annual surprise)	Fourth-quarter EPS minus fourth-quarter EPS a year earlier, divided by the SD of the firm's last eight such errors published before the report (at least four).
SUE^Q (Q1 surprise)	The same for the Q1 report of the following year.
Same day / Bundle	Annual and Q1 report published on the same date.
Same month	Both reports in the same calendar month, on different days.
Separate months	The Q1 report in a later month than the annual report.
Lateness	Within-year percentile of the annual-report date; days after the fiscal year end.
Return	Monthly return minus the equal-weighted return of the stock's size decile.
Abnormal turnover	Log monthly turnover minus the firm's mean over months -12 to -2.
Size	Log market value at the start of the annual-report month.
Analyst coverage	I/B/E/S 12-month-forward consensus available in the annual-report month.

Replication. All results come from the Python scripts in the folder `implementation/code`; `run_all.py` runs them in order. Script `s00` builds the sample, and scripts `s01` to `s06` produce the tables, the figures and every number quoted in the text, which they write as LaTeX macros, so the text and the tables cannot diverge.